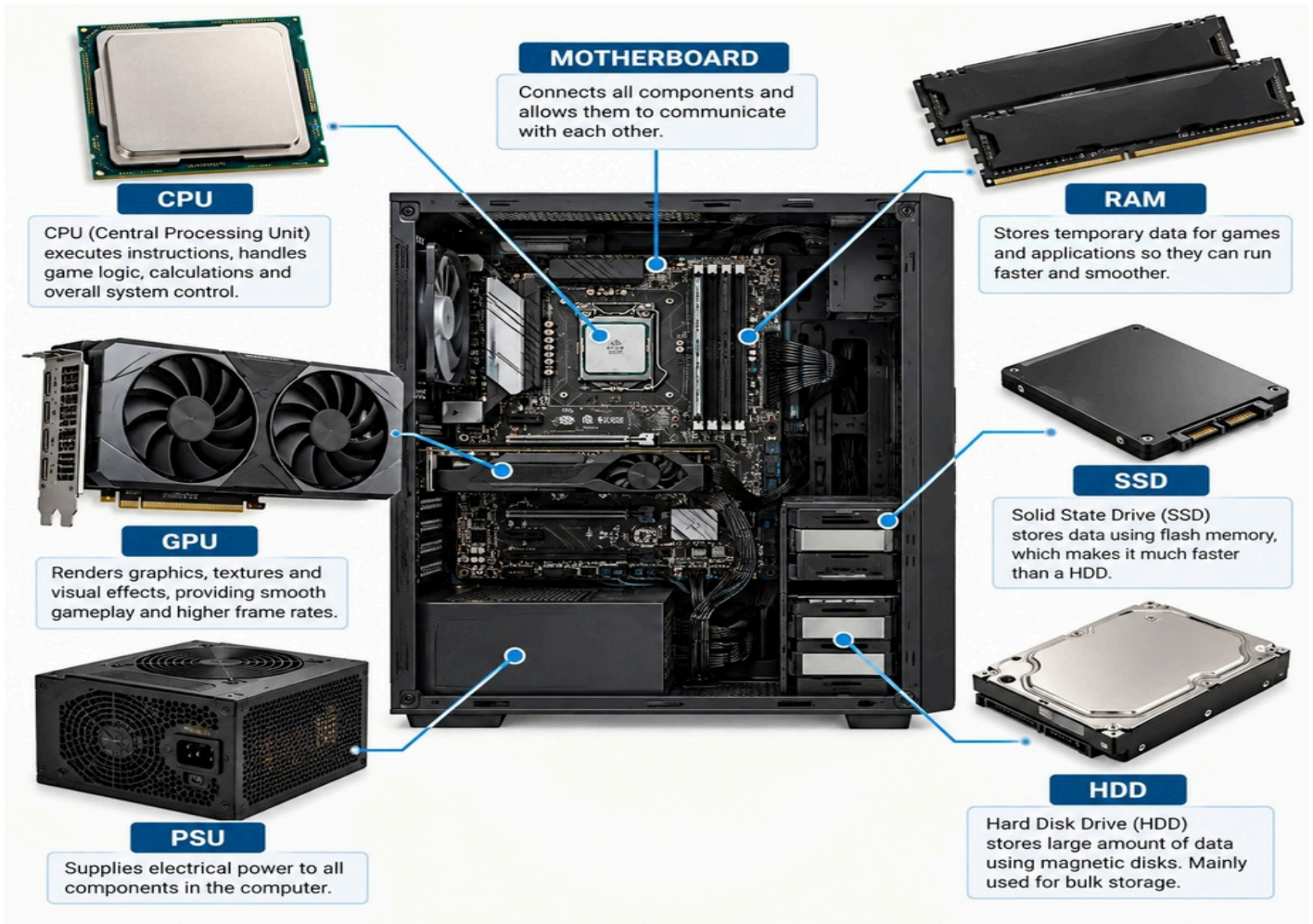


ASSIGNMENT 1.1 — TASK 1

Labelled desktop PC hardware diagram showing the motherboard, CPU, RAM, GPU, SSD, HDD and power supply.



Task answer: The diagram identifies the main internal components of a desktop PC and shows their relationship to the motherboard. The CPU performs general processing, RAM provides temporary working memory, the GPU handles graphics rendering, the SSD/HDD provide permanent storage, and the PSU supplies electrical power to the system.

Task 2

Compare a Hard Disk Drive (HDD) and a Solid-State Drive (SSD). Focus on the factors that matter when choosing storage for a desktop PC.

Technology	Stores data on magnetic platters with moving mechanical parts.	Stores data in flash memory chips with no moving storage mechanism.
Speed	Generally slower, especially for random access and application loading.	Generally much faster for booting, loading applications and accessing files.
Durability	Mechanical parts make it more sensitive to shocks and physical movement.	No moving parts make it better suited to handling everyday movement and vibration.
Power use	Usually consumes more power during operation because the drive mechanism must move.	Usually uses less power for comparable workloads.
Noise	Can produce audible spinning or seeking sounds.	Operates silently.
Typical use	Large-capacity storage, backups and budget-oriented systems.	Operating-system drive, gaming, software and fast everyday storage.
Best reason to choose	Useful when capacity per unit cost is the main consideration.	Useful when speed, responsiveness and durability are priorities.

Summary: HDDs remain useful when large storage capacity and cost are important, while SSDs are normally preferred when quick boot times, faster game/application loading and a more responsive system are the main goals.

Task 3

Explain how the CPU, RAM and GPU work together when running a game such as PUBG or Free Fire on a PC.

CPU — Game Control & Calculations

The CPU processes game instructions such as game rules, player movement, physics calculations and background tasks. It coordinates many of the decisions that must happen while the game is running.

RAM — Active Game Workspace

RAM temporarily holds data that the game and operating system need while playing. Adequate RAM helps keep active game data readily available and reduces slowdowns caused by insufficient memory.

GPU — Graphics Rendering

The GPU converts game data into the images displayed on the monitor, including characters, environments, lighting, textures and effects. A capable GPU helps maintain smoother frame rates and allows higher graphics settings.

Task 4

Imagine upgrading a desktop PC so that a modern FIFA title can run smoothly. Identify the components you would consider upgrading and explain the reason for each.

GPU	The graphics card performs the majority of real-time 3D rendering work. If the current GPU is weak, it can limit graphics quality and frame rate.	Higher and more stable FPS, improved image quality and support for higher graphics settings.
RAM	Additional memory gives the game and operating system more working space. This is useful when the current memory capacity is insufficient.	Fewer memory-related slowdowns and better multitasking while gaming.
CPU	The processor handles game logic, simulation, player behaviour and other calculations. A weak CPU can become a performance limit even when the GPU is capable.	Better game simulation, more consistent frame delivery and improved performance in CPU-heavy situations.
SSD	Moving the operating system or game from a slower drive to an SSD improves storage responsiveness.	Faster booting, quicker game/application loading and shorter loading screens.

Upgrade priority

GPU: This is a major consideration for a gaming-focused upgrade because it directly affects 3D rendering, graphics quality and frame rate. The CPU and RAM should also be checked so that they do not restrict the GPU. An SSD improves loading and general responsiveness rather than directly increasing graphics FPS.

Final observation

A balanced gaming PC is more useful than upgrading only one component without considering the rest of the system. The best upgrade depends on the current hardware, the game's requirements, target resolution, desired graphics settings and available budget.